

DHRUV VASANIYA

Electronics & Telecommunication Engineering Student | Hardware & Embedded Systems

vasaniyadhruv2006@gmail.com • +91 7738919738 • Mumbai, Maharashtra, India
linkedin.com/in/dhruv-vasaniya • github.com/dhruvvasaniya07

PROFESSIONAL SUMMARY

Second-year Electronics and Telecommunication Engineering student at VIVA Institute of Technology (Mumbai University) with hands-on experience in hardware design, robotics, and IoT systems. Multiple award-winning project developer with 1st Prize at J.P. Morgan & Agastya Foundation School Innovation Challenge. NPTEL Elite certified in Experimental Robotics (IIT Kharagpur, 85%) and Oracle Cloud Infrastructure AI Foundations certified. Passionate about innovative hardware solutions and emerging technologies in electronics and telecommunications.

EDUCATION

Bachelor of Engineering (BE) — Electronics and Telecommunication Engineering 2024 – 2028

VIVA Institute of Technology, Mumbai University

- Currently in Second Year
- Focus: Electronics, Telecommunications, Hardware Design, Embedded Systems

CERTIFICATIONS

NPTEL Elite — Experimental Robotics Feb – Mar 2026

IIT Kharagpur (NPTEL / Swayam, Govt. of India)

- Score: 85% | Online Assignments: 25/25 | Proctored Exam: 60/75
- Skills: Robotics Fundamentals, Experimental Methods, Embedded Systems, Problem Solving

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate 2025

Oracle

- Skills: Cloud Architecture, Infrastructure Management, AI/ML Fundamentals, Data Analytics

PROJECTS & ACHIEVEMENTS

Smart Attendance System ■ 1st Prize

Agastya Foundation & J.P. Morgan — The School Innovation Challenge, Mumbai

- Designed and built a Smart Attendance System prototype for real-world deployment
- Handled end-to-end hardware setup — circuit design, wiring, troubleshooting, and final integration
- Role: Hardware & Software Lead — ensured system worked in real-life conditions, not just on paper
- Technologies: Embedded Systems, Hardware Design, Software Integration, IoT, Prototyping

Team: Dhruv Vasaniya & Yash Patil | Mentor: Nutan Malekar

Fire-Fighting Robot ■ 3rd Consolation Prize

Anveshana Mumbai 2025–26 Science & Engineering Fair — Agastya International Foundation × Samsung Semiconductor India Research

- Designed low-cost autonomous fire-fighting robot to address agricultural fire hazards
- Implemented early detection and targeted fire suppression system
- Reduced crop loss and optimized water usage through wireless communication technology
- Technologies: Robotics, Autonomous Systems, Wireless Communication, IoT, Fire Detection

Team: Dhruv Vasaniya & Yash Patil

Electric Whistle ■■ Appreciation Award — Best Presentation

Agastya Foundation & J.P. Morgan — School Innovation Challenge

- Built a smart, practical Electric Whistle solution from concept to functional prototype
- Role: Full-Stack & Integration Lead — handled software side, connected hardware logic to functional system
- Delivered high-impact presentation that resonated with the industry panel
- Technologies: Full-Stack Integration, Hardware Logic, Embedded Systems, Prototyping

Team: Dhruv Vasaniya & Yash Patil

Fire & Smoke Detection System

Convergence 2025 — IETE-VIVA Inter-College Competition | Presented at Oscillations 2025 (IETE Mumbai)

- Developed Arduino-based real-time fire and smoke detection system
- Integrated MQ7 sensor, flame sensor, LCD display, and buzzer for instant safety alerts
- Overcame last-minute technical challenges through debugging and teamwork
- Technologies: Arduino, MQ7 Sensor, Flame Sensor, LCD Display, Embedded C

Team: Dhruv Vasaniya, Yash Patil, Riddhi Thakare & Sanchitha Poojari

TECHNICAL SKILLS

Electronics & Telecommunication:

Analog Circuit Design, Digital Electronics, Microcontrollers & Embedded Systems, Signal Processing

Cloud & AI Technologies:

Oracle Cloud Infrastructure (OCI), AI/ML Fundamentals, Cloud Computing Concepts, Data Analytics

Hardware & Design Tools:

PCB Design & Layout, Arduino, Raspberry Pi

Communication Technologies:

Wireless Communication, Network Protocols, IoT Systems, RF Design Basics

Programming:

C/C++ (Embedded), Python, VHDL / Verilog, Assembly Language, Version Control (Git)

CURRENTLY LEARNING

- Advanced PCB Design — Multi-layer PCB design and RF circuit layout
- Machine Learning for IoT — Edge AI and TinyML applications
- 5G Communication Systems — Next-generation wireless protocols

PROFESSIONAL HIGHLIGHTS

- ✓ Multiple award-winning projects validated by national-level competition success
- ✓ Oracle certified in Cloud & AI technologies
- ✓ NPTEL Elite certified in Experimental Robotics from IIT Kharagpur with 85% score
- ✓ Hands-on experience in building and deploying real hardware systems
- ✓ Strong problem-solving and quick learning abilities demonstrated under competition pressure
- ✓ Collaborative team player with consistent successful project delivery across different domains

AREAS OF INTEREST

Robotics • Internet of Things (IoT) • Embedded Systems • Hardware Innovation • Wireless Communication • Edge AI • Renewable Energy Systems • Smart Agriculture